

Research



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Animal behaviour

Maternal effects impact decision-making in a viviparous lizard

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Stressful conditions experienced during early development can have deleterious effects on offspring morphology, physiology and behaviour. However, few studies have examined how developmental stress influences an individual's cognitive phenotype. Using a viviparous lizard, we show that the availability of food resources to a mother during gestation influences a key component of her offspring's cognitive phenotype: their decision-making. Offspring from females who experienced low resource availability during gestation did better in an anti-predatory task that relied on spatial associations to guide their decisions, whereas offspring from females who experienced high resource availability during gestation did better in a foraging task that relied on colour associations to inform their decisions. This shows that the prenatal environment can influence decision-making in animals, a cognitive trait with functional implications later in life.

1. Introduction

Cognitive processes allow animals to perceive, consolidate and act on information acquired during development—playing a fundamental role in an organism's ability to address environmental challenges [1,2]. Despite this, our understanding of what drives individual variation in cognitive traits is limited [3]. As a consequence, we are ill-equipped to understand the role intra-individual variation in cognition plays in mediating evolutionary and ecological processes.

It has long been suggested that stressors experienced in early life impact offspring phenotypic traits, often with profound and long-term effects [4,5]. Indeed, the early developmental environment affects a range of morphological, behavioural and physiological traits [6]. Early life stressors have also been suggested to shape an individual's cognitive phenotype owing to differential or impaired allocation of resources to brain formation [7]. Despite this, few empirical studies have examined early environmental effects on the development of cognitive traits themselves (but see [1]).

Here we use a viviparous lizard to examine how food availability during two periods early in development, pre- and post-birth, affects offspring decision-making and learning. To achieve this, we subjected offspring raised on low or high resource availability to two cognitive tasks, a 'foraging task' that relied on colour associations and an 'anti-predatory task' that relied on spatial associations.

2. Material and methods

(a) Animal collection and experimental design

We collected 70 pregnant female *Liopholis whitii* from Orford, Tasmania, at the start of gestation. Females were transported to the University of Tasmania, measured for

morphometric traits (weight (± 1 mg) and snout–vent–length (SVL; ± 0.5 mm)) and housed in outdoor enclosures (1 m diameter). Females were then randomly allocated to one of two pre-birth resource treatments. Females in the low resource treatment were provided one mealworm (*Tenebrio molitor*) three times weekly, whereas females in the high pre-birth treatment received five mealworms three times weekly. These treatments resulted in significant differences in female body condition at the end of gestation (analysis of covariance, $F_{1,65} = 68.60$, p -value < 0.001).

At the end of gestation (mid-January) females were moved to indoor terrestrial ecology facilities to give birth. At birth, we recorded offspring morphometric traits. We were unable to identify offspring sex as juvenile female *Liopholis* retain their hemipenes until sexual maturity. Females and their offspring were then randomly assigned to a post-birth treatment that was either the same or the opposite of the one the female experienced during gestation. In the post-birth treatment, we doubled the number of mealworms provided to account for the presence of both the female and her offspring. Females and their offspring were released into the small outdoor enclosures and kept under the post-birth resource treatments for a six-week period. At the completion of this six-week period, offspring were returned to the laboratory where they were remeasured for morphometric traits and housed individually. We then assayed all offspring for two cognitive tasks.

(b) Cognitive tasks

We subjected offspring to two cognitive tasks to test for differences in their ability to solve these tasks and to test whether differences were dependent on the context of the task and the cues available to inform decisions. The order of assays was randomized across all offspring. For both tasks, offspring were assayed twice daily (morning and afternoon) over 10 consecutive days ($n = 20$ trials per cognitive task). We scored the offspring's first choice (correct versus incorrect). All assays were scored blind to the offspring's treatment.

(i) Foraging task

Offspring had to learn to associate a food reward with a specific coloured block [8]. For each trial, we placed a food dish on each of two elevated blocks (7 (L) $\times$ 7 (W) $\times$ 4 (H) cm), one blue and one white in front of the offspring such that the food could not be seen. One block allowed access to a mealworm, while access to the mealworm on the other block was prevented by a mesh screen inside the dish. We gave the offspring a maximum of 1 h to attempt the task. We considered the offspring to have made a choice if it placed both its forelimbs on the top edge of the block. The colour and position (right or left) of the correct block was randomized and counter-balanced across treatment groups to account for colour or spatial bias between offspring.

(ii) Anti-predatory task

We set up a biologically relevant anti-predatory paradigm, which has been used with success in previous studies [9]. In this task, offspring had to learn the location of a 'safe' refuge when given the option of two refuges. Specifically, we simulated predatory attacks using a standardized protocol (tapping the offspring from behind on the pelvic girdle using an index finger) allowing the offspring to choose its flight direction until it entered the 'safe' refuge. If it entered the 'unsafe' refuge, we lifted the refuge and resumed chasing (the refuge was immediately replaced in its original position). We gave the offspring a maximum of 10 min to complete the task. We considered the offspring to have made a choice when it ran inside a refuge. We kept the location of both refuges constant throughout the trials to determine whether offspring use spatial cues to solve the task but

randomized the location of the 'safe' refuge between offspring within each resource treatment.

(c) Statistical analysis

We analysed differences in decision-making (overall probability of choosing correctly) and learning rates (increase in the probability of choosing correctly across trials) using generalized linear mixed models in the lme4 package [10] in R v. 3.0.3 [11]. Within our models, we included pre- and post-birth treatment and their interaction, z-transformed trial and z-transformed SVL at six weeks as fixed factors and individual and maternal identity as random effects. We also included an interaction between treatment and trial to test for differences in learning rates and considered an increase in the probability of making correct choices across trials as evidence for learning. The inclusion of a random slope (trial) in our models led to poor model convergence and was dropped from subsequent analyses. Model reduction was performed following the backward elimination of non-significant interactions ($p > 0.05$) and resulted in main effects models.

We tested for lateralization bias and none was found. See the electronic supplementary material for full details on methodology and statistical analyses.

3. Results

We found a significant effect of pre-birth resource treatment on the probability of choosing correctly in both tasks; however, the direction of the effect was opposite for the two tasks (table 1). Offspring from the low pre-birth treatment had a higher than expected probability of choosing correct in the anti-predatory task and differed significantly from offspring developing under high pre-birth resources. This pattern was reversed in the foraging task, where offspring from the high pre-birth treatment had a higher than expected probability of choosing correct—differing significantly from offspring developing under low resources (figure 1). We found no evidence that the probability of offspring choosing correctly increased across trials, indicating that the cognitive process affected was decision-making rather than learning the tasks *per se*. We found some evidence that this lack of learning was a result of over-training or changed motivation across trials (i.e. modelling a nonlinear—quadratic—trial effect) in the foraging task, but found no evidence for this in the spatial task (see the electronic supplementary material). We found no effect of the post-birth resource treatment on the probability of choosing correctly in either task (table 1).

4. Discussion

Our results suggest that the resource environment a mother experiences during gestation influences offspring decision-making. Offspring exposed to low prenatal resource availability did better in the anti-predatory task with spatial associations to guide decisions, while offspring that developed under high prenatal resource availability did better in the foraging task with colour associations as guidance. These results were independent of any clear evidence of learning to solve these particular tasks, suggesting that the prenatal environment influenced decision-making patterns rather than task-specific learning.

The absence of learning in these tasks is surprising given that previous studies on adult *L. whitii* have been shown to

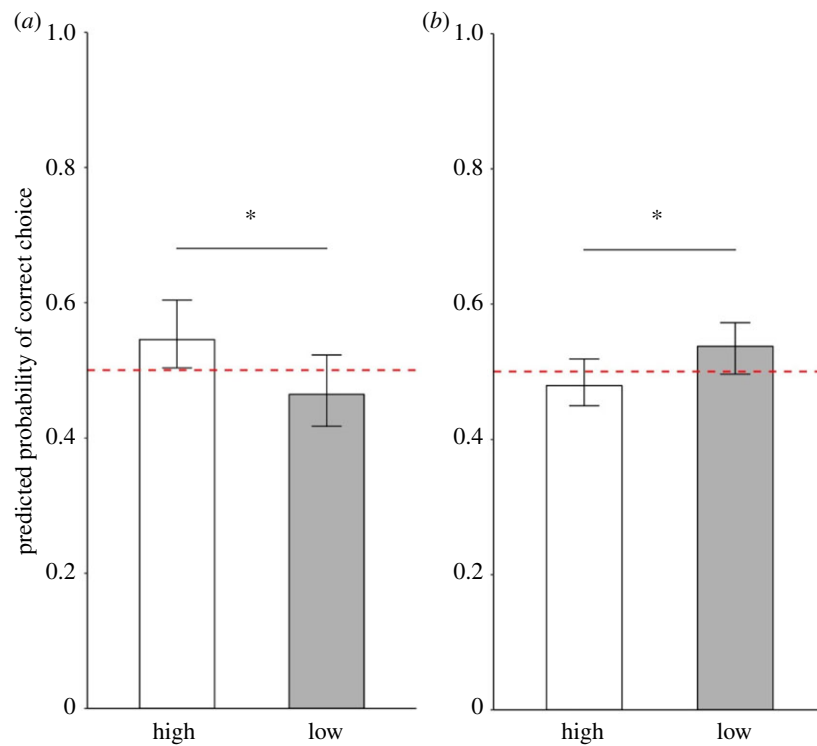


Figure 1. Mean ($\pm$ 95% CI) predicted probability of choosing correct in (a) a foraging and (b) an anti-predatory task for offspring exposed to high or low pre-birth resource availability. The dashed line represents 50/50 chance. Asterisk (*) denotes differences significant at $p < 0.05$. (Online version in colour.)

Table 1. Parameter estimates and 95% CI of generalized linear mixed models examining the relationship between pre- and post-birth resource availability on offspring's probability of choosing correctly in a foraging and an anti-predatory task. Statistically significant p -value values are in *italics*. Main effects are presented from a model without interactions.

	foraging task				anti-predatory task			
	estimate (β)	lower CI	upper CI	p -value	estimate (β)	lower CI	upper CI	p -value
intercept	0.25	−0.03	0.54	0.07	−0.05	−0.22	0.12	0.57
scaled SVL	0.21	0.05	0.38	0.01	−0.10	−0.21	0.01	0.08
scaled trial number	0.07	−0.04	0.19	0.22	0.02	−0.09	0.12	0.73
treatment pre-birth LOW	−0.41	−0.77	−0.05	0.02	0.25	0.02	0.47	0.03
treatment post-birth LOW	0.007	−0.33	0.36	0.97	−0.07	−0.28	0.14	0.52
scaled trial $\times$ pre-birth	0.05	−0.18	0.28	0.67	0.11	−0.10	0.32	0.30
scaled trial $\times$ post-birth	−0.02	−0.25	0.21	0.85	0.002	−0.21	0.21	0.30
pre- $\times$ post-birth	0.03	−0.67	0.72	0.92	0.001	−0.57	0.29	0.99

them capable of learning both cognitive tasks (KL Munch, DWA Noble, E Wapstra & GM While 2018, unpublished data). While changed motivation or over-training may explain a lack of learning in the foraging task, it remains unclear why no learning was detected in the anti-predatory task. There are a number of potential explanations for this, including age-specific constraints [12] or an inability to cope with stress, which is known to affect learning [13]. Alternatively, learning may have occurred but in more nuanced ways that we failed to pick up with our experimental design. Indeed, the nonlinear relationship between probability of correct choice and trial number in the foraging tasks suggests more subtle effects on learning may be occurring. More data on how developmental conditions influence learning *per se* are required to tease apart these alternative explanations.

Irrespective of the lack of task-specific learning, the broad effect of the prenatal environment on the development of

cognitive traits in *L. whitii* is in agreement with effects in other taxa. Indeed, studies have shown that a range of pre-natal conditions (e.g. malnutrition, hormone exposure) can have substantial impacts on the development of key traits associated with an individual's cognitive ability (e.g. learning and memory, see [1] for review). For example, geckos incubated in hot temperatures had poorer spatial learning ability, than geckos incubated in colder temperatures [14]. However, despite a consistent overall effect of prenatal conditions impacting the development of decision-making, the relative direction of these effects differed depending on the cognitive task, with offspring experiencing poor resource conditions during gestation making more correct decisions in the anti-predatory task but worse in the foraging task. One explanation for these results is that natural selection has shaped the developmental trajectory of decision-making in relation to the likely resource environments offspring

find themselves in post-birth (i.e. a context-dependent anticipatory maternal effect [15]). More work targeting whether the subtle changes we see in decision-making between treatments translate into fitness benefits under different postnatal conditions is required to tease apart these explanations. Alternatively, these effects may be a non-adaptive consequence of differential resource allocation to different areas of the brain. In reptiles, the spatial domain is linked to the medial cortex while visual information (such as colour) is processed in the dorsal cortex [16]. Therefore, any trade-off between resource allocation to these areas of the brain for other functional reasons may feedback to influence decision-making. The integration of functional outcomes of cognitive traits with a fundamental understanding of the neural biology underlying those traits is lacking for the majority of non-model systems but will provide a fruitful avenue for future research.

We found no evidence for effects of postnatal environment on the probability of choosing correctly in either cognitive task. As all offspring were held with their mothers in the postnatal environment we cannot rule out potential carry-over effects of the prenatal environment on maternal behaviour that may, in turn, have masked any effects of the postnatal environment. However, the strong effects of the postnatal environment on other aspects of offspring

development (e.g. improved offspring SVL growth; see the electronic supplementary material) suggest that the lack of effects is likely to be real.

In summary, we found that decision-making is affected by prenatal resource conditions. While our data do not provide insight into the mechanism(s) responsible or the fitness consequences, they suggest that early life stressors may evoke a trade-off between allocating resources to developing different cognitive domains, enhancing one at the expense of another.

Ethics. Research approved by the University of Tasmania Animal Ethics Committee (A0016084).

Data accessibility. The dataset is available at the Dryad digital repository: <https://dx.doi.org/10.5061/dryad.01vc6> [17].

Authors' contributions. K.L.M., D.W.A.N., T.B.J., E.W. and G.M.W. conceived the study. Data were collected by K.L.M., I.S.K., B.H. and T.B.J. K.L.M., D.W.A.N. and G.M.W. performed the analysis. All authors wrote the manuscript, approved the version to be published and agreed to be accountable for all aspects of the research.

Competing interests. We have no competing interests.

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